NETWORK DEVICES & SELECTION LAB

1. Research two actual network switches available in India (for example, from Cisco, TPLink, or D-Link) and list their key specifications such as number of ports, speed, and price in a table.

Answer & Specifications:

Brand & Model	Number of Ports	Port Speed	Approx. Price (INR)
Cisco Catalyst 1000 Series (C1000-24T-4G-L) [cite: 1.1.4]	24 Gigabit RJ45 + 4 SFP Uplinks [cite: 1.1.4]	10/100/1000 Mbps (Gigabit) [cite: 1.1.4]	₹35,000 - ₹45,000 [cite: 1.1.4, 2.1.1]
TP-Link JetStream (TL-SG3428) [cite: 1.2.2]	24 Gigabit RJ45 + 4 SFP Slots [cite: 2.2.2]	10/100/1000 Mbps (Gigabit) [cite: 2.2.2]	₹11,000 - ₹13,500 [cite: 1.2.2, 2.2.3]

2. Create a device selection matrix comparing at least one switch, one router, and onefirewall that could be used for a small startup office (20-30 employees). Include columns for brand/model, key specs, price, and your rating for each device.

Answer & Selection Matrix:

Device Type	Brand / Model	Key Specifications	Approx. Price	Rating
Switch	TP-Link JetStream TL-SG3428 [cite: 1.2.2]	24 Gigabit ports, 4 SFP slots, L2+ Managed, Omada SDN [cite: 1.2.2, 2.2.2]	₹12,000 [cite: 2.2.3]	4.5 / 5
Router	MikroTik hEX (RB750Gr3)	5 Gigabit ports, Dual-core 880MHz CPU, IPsec hardware encryption	₹5,500	4.4 / 5
Firewall	Fortinet FortiGate 40F	5 Gigabit ports, Next-Gen Security, SSL VPN throughput 900 Mbps	₹45,000	4.8 / 5

3. Imagine you are setting up the network for a new gaming café with 25 computers andpublic Wi-Fi. Select one switch, one router, and one firewall model for this scenario and justify your choices in 2-3 lines each.

Answer & Justifications:

- **Switch: TP-Link TL-SG3428 (24-Port Managed Switch):** [cite: 1.2.2] Provides 24 gigabit ports to connect the gaming PCs directly while supporting VLAN segmentation to isolate guest Wi-Fi traffic from high-performance gaming terminals. [cite: 1.2.2, 2.2.2]
- **Router: MikroTik CCR1009-7G-1C-1S+:** Delivers heavy multi-gigabit routing performance and robust bandwidth management to handle high-speed internet required by 25 concurrent competitive gamers without latency spikes. [cite: 1.2.1]
- **Firewall: Fortinet FortiGate 60F:** Offers advanced threat protection, application control, and captive portal features for secure public Wi-Fi access while ensuring secure NAT and DDoS prevention for the café.

4. Prepare a Bill of Materials (BOM) listing the selected network devices (switch, router,firewall) for your gaming café scenario, including quantity, model, unit price, total price, and a short justification for each item.

Answer & Bill of Materials (BOM):

Item	Qty	Model	Unit Price (INR)	Total Price (INR)	Justification
Switch	1	TP-Link TLSG3428 [cite: 1.2.2]	₹12,000 [cite: 2.2.3]	₹12,000	Ample Gigabit ports for 25 PCs + uplinks [cite: 2.2.2]
Router	1	MikroTik CCR1009 [cite: 1.2.1]	₹32,000	₹32,000	High throughput & packet forwarding for gaming [cite: 1.2.1]
Firewall	1	Fortinet FortiGate 60F	₹65,000	₹65,000	Robust UTM firewall & captive portal for public Wi-Fi
Total Estimated Investment				**₹109,000**	

5. Perform a cost-benefit analysis for your chosen devices: explain in 3-4 lines why youselected these models over cheaper or more expensive alternatives, considering both features and budget.

Answer & Cost-Benefit Analysis:

The chosen hardware stack balances high performance and cost-effectiveness by avoiding expensive enterprise-tier Cisco core switches while rejecting unmanaged consumer gear that lacks security. The TP-Link JetStream switch and MikroTik router offer advanced L2+ management and enterprise routing features at a fraction of enterprise brand costs. Meanwhile, investing in a dedicated FortiGate firewall ensures robust security, guest Wi-Fi isolation, and DDoS protection, safeguarding business uptime without overspending on datacenter-grade hardware.